

Identifying Risk Factors for Postpartum Depression

Leakage-controlled classification using Mendeley Data

Jon Steen | Bellevue University
DSC680 | MILESTONE 3

Support earlier screening, not diagnosis

Business problem

Elevated risk can go unrecognized without timely screening.

Behavioral health and outreach capacity may be limited.

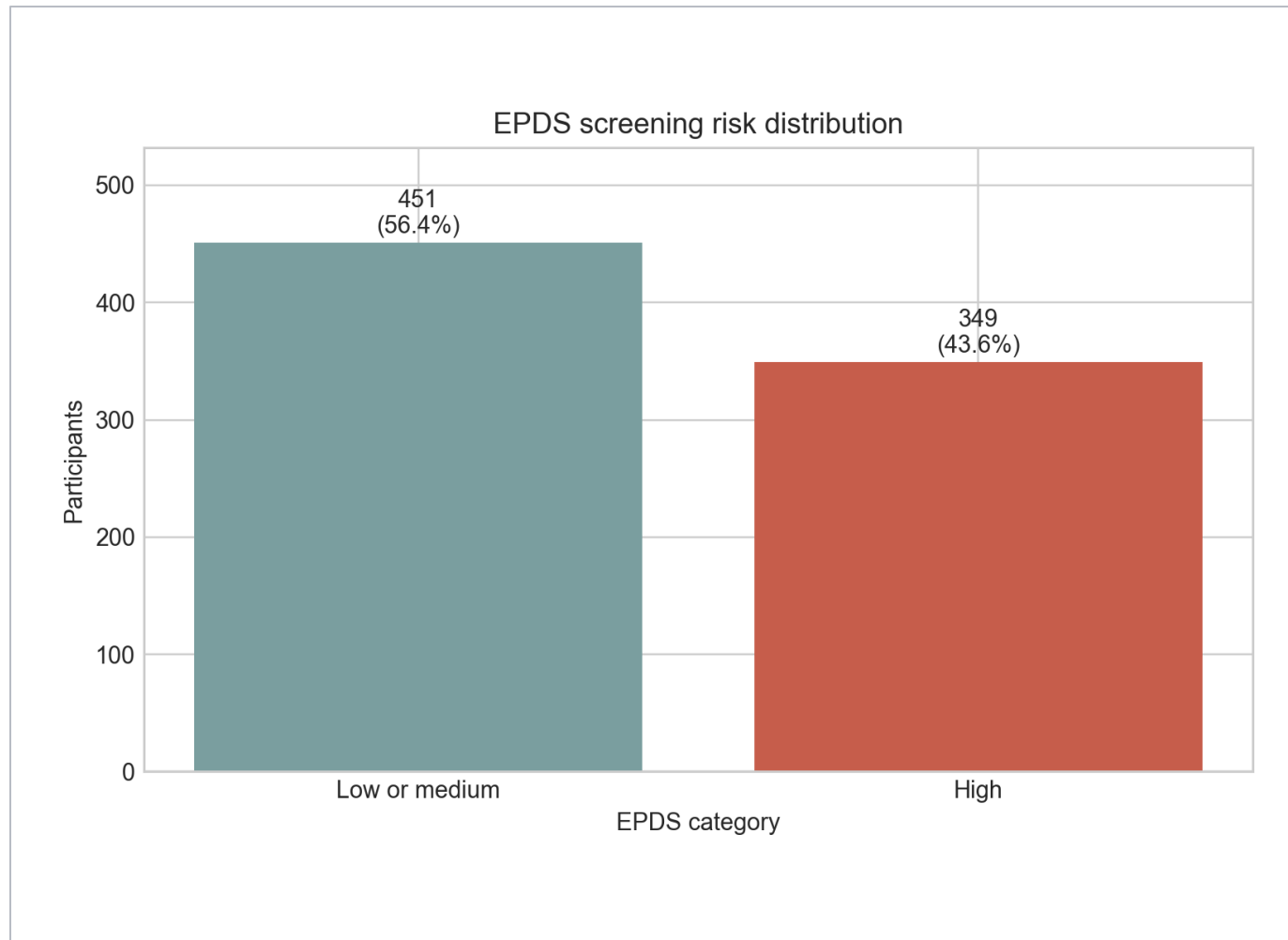
A useful tool should direct attention toward validated screening and follow-up.

Analytical question

Can routinely available, non-symptom characteristics identify elevated EPDS screening risk?

Use case: risk-based outreach that preserves standardized screening and clinical review.

Strict leakage control preserves validity



800 participants

349 high risk | 43.6%

70 raw columns and 38 retained predictors
before encoding

EPDS and current PHQ-9 items removed

Selected postpartum symptom proxies
removed

206 ambiguous blanks retained as an
explicit missing level

The target is screening risk, not diagnosis.

Validation separates development from test

All preprocessing was learned inside each training fold.

800

total rows

Stratified split with seed 680

600

development rows

Repeated 5-fold CV, 5 repeats

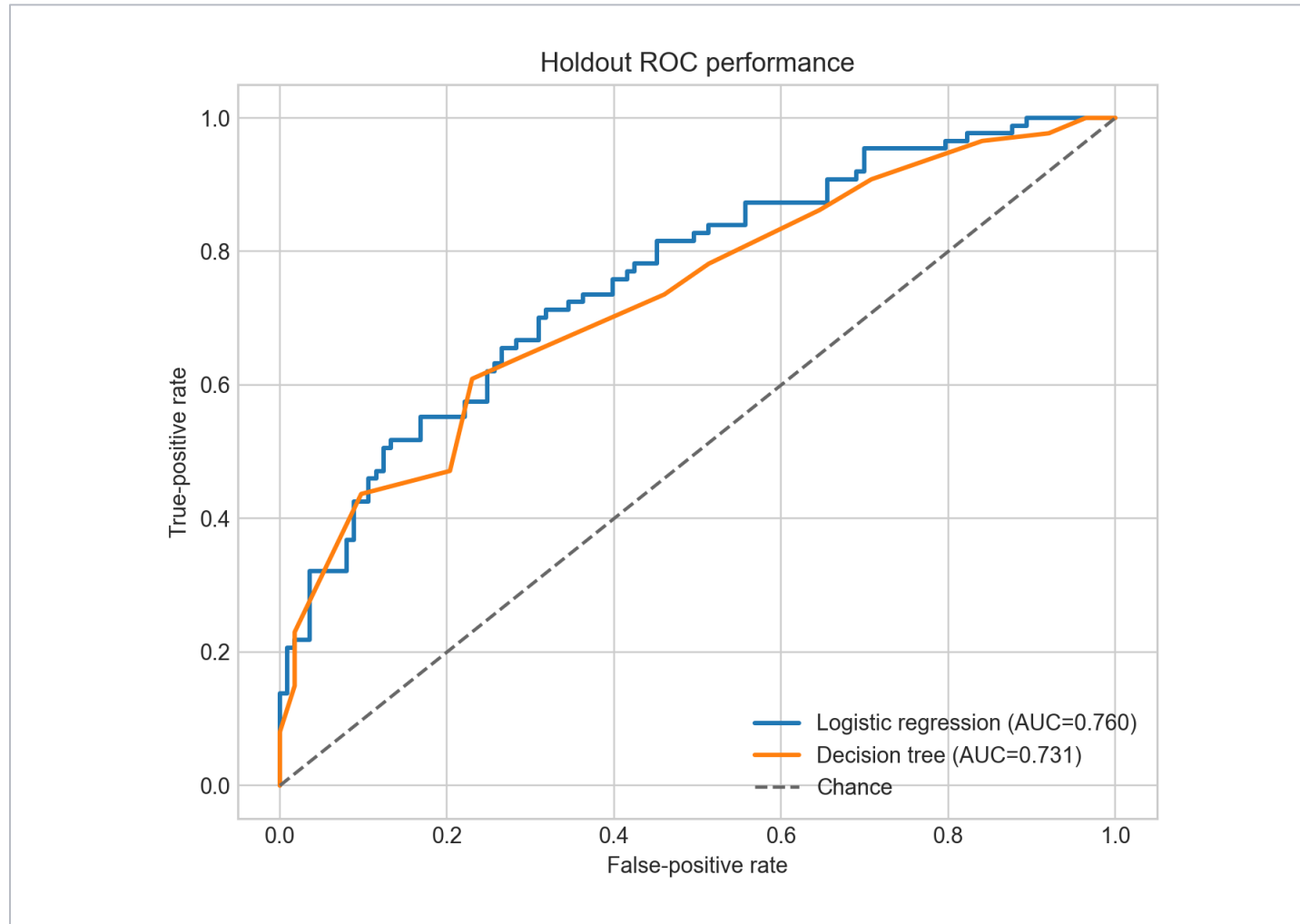
200

holdout rows

Used once for final comparison

Compared models: regularized logistic regression and a shallow decision tree

Logistic regression was the stable baseline



0.774

mean CV ROC-AUC

0.760

holdout ROC-AUC

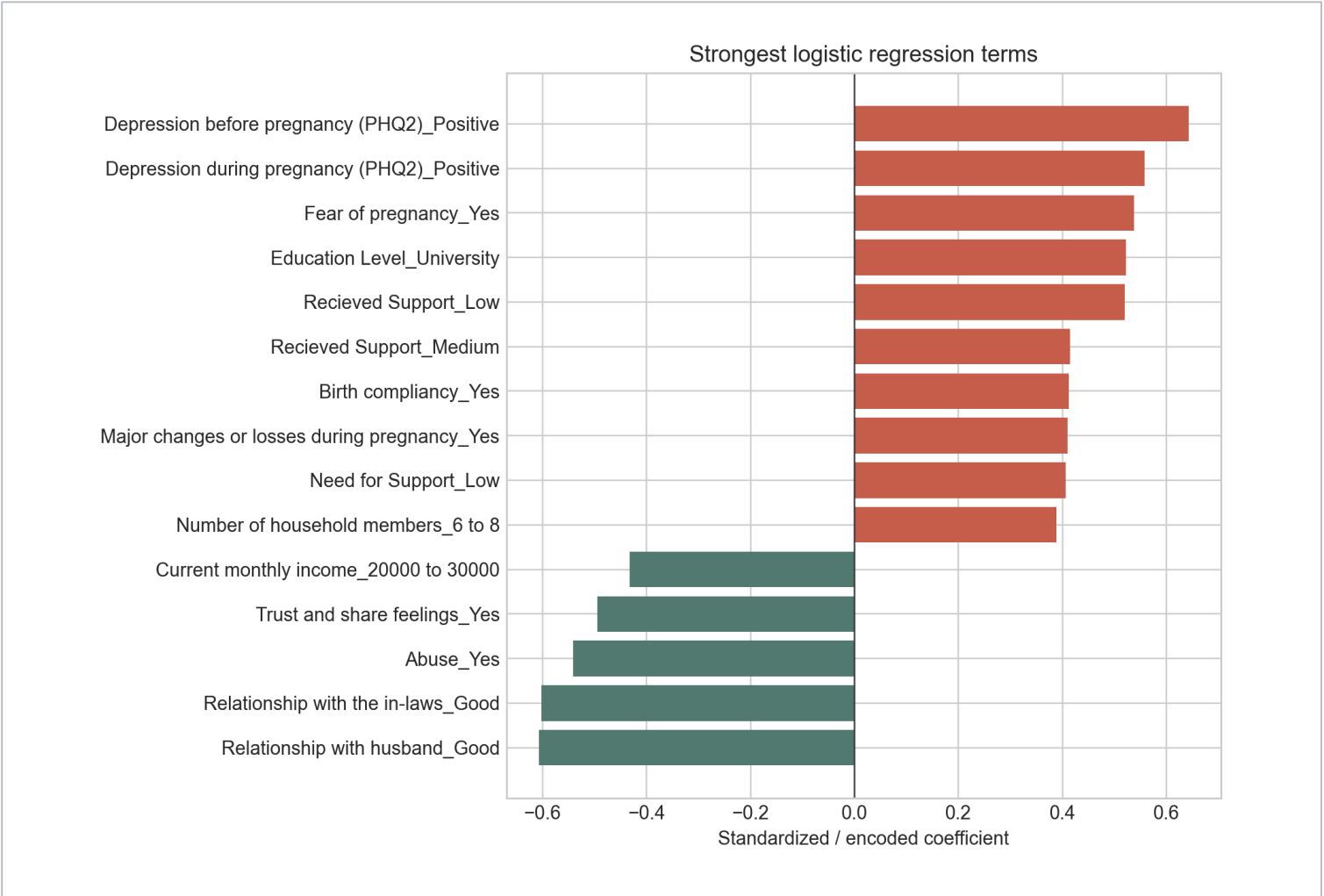
Logistic: recall 0.598, precision 0.650
Tree: recall 0.736, precision 0.552
Threshold choice controls the tradeoff

Prior mental health and support drove signal

Strongest positive terms

- Depression before pregnancy
- Depression during pregnancy
- Fear of pregnancy
- Lower reported support
- Birth complications and major losses

Predictive association
not causal effect



Local evidence limits current use

Limitations

Bangladesh sample with limited U.S. generalizability
Cross-sectional, self-reported data
Postpartum timing ranges up to 24 months
Small sample and substantial missingness

Ethical boundaries

Protect sensitive mental health and family data
Measure false-negative and false-positive harms
Audit subgroup calibration and disparities
Prohibit punitive, insurance, custody, and employment uses

Local prospective validation is required before operational use.

Six phases protect patients before rollout

Each stage creates evidence and accountability before the next stage begins.

01-02

Governance + local data

Define intended use, privacy, consent, ownership, escalation, predictor timing, and outcome collection.

03-04

Validation + silent pilot

Test discrimination, calibration, subgroups, threshold policies, alert volume, and workflow burden.

05-06

Limited rollout + monitoring

Prioritize outreach only. Track drift, errors, disparities, referrals, and patient complaints.

FINAL RECOMMENDATION

Use risk scoring only to prioritize validated screening and follow-up.

Prospective local validation is the next evidence requirement.